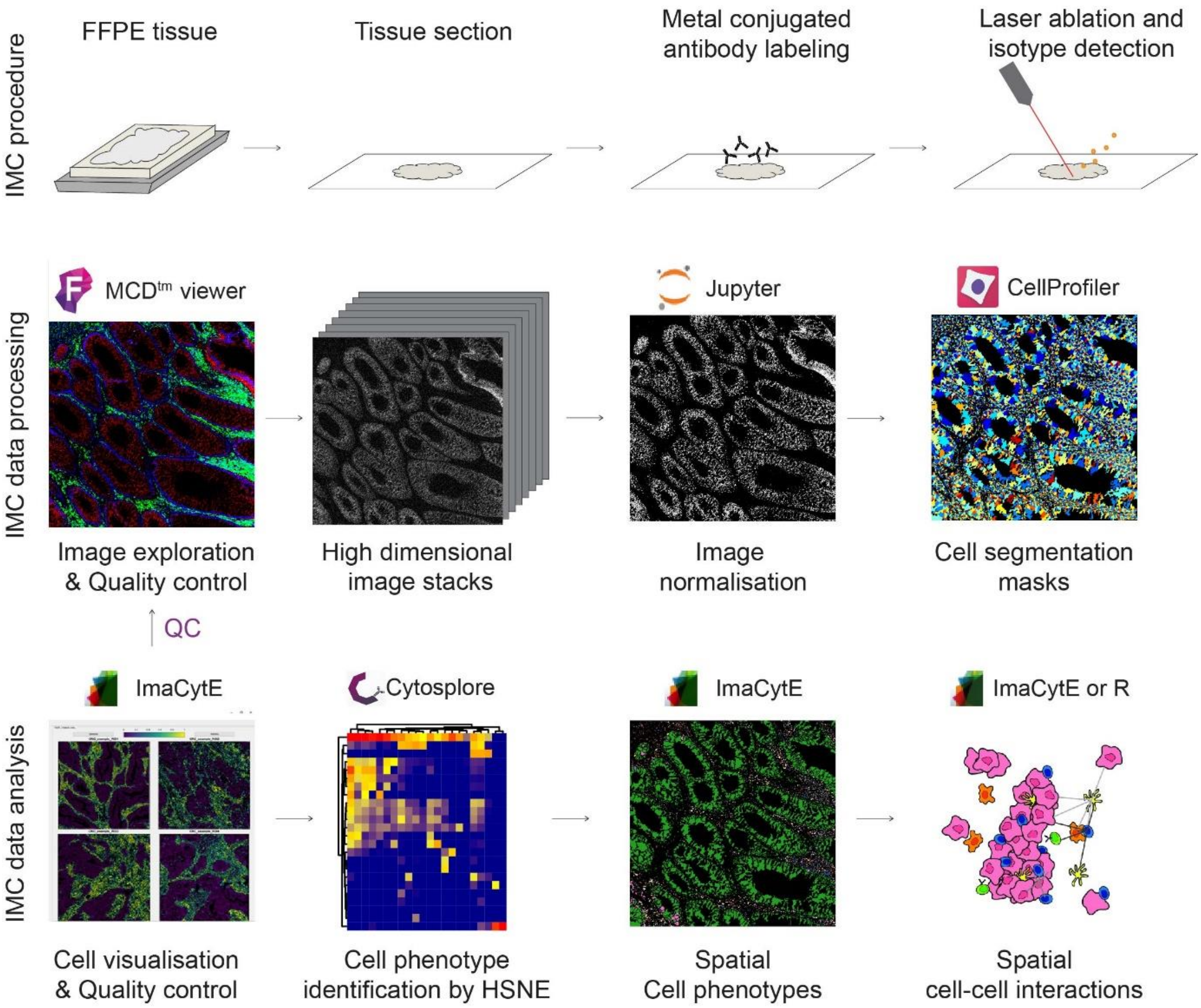
Imaging mass cytometry data analysis v10

Overview of the pipeline



Software to download

- MCD-viewer (any version)

  Used to view MCD files, file conversion and thresholding.

  Software is only available for windows and administrator rights are needed to open. Can be downloaded from www.fluidigm.com/software

- Penguin

  Used for image normalisation

  Available on github https://github.com/deMirandaLab/PENGUIN including installation instructions

- Cellprofiler

  Used to convert probability masks from ilastik to masks

  Can be downloaded from https://cellprofiler.org/previous_releases/

  Software needs java 64-bit to install which is only installed if the browser used to download is 64-bit.

- Imacyte

  Used for the analysis of images, phenotype identification, neighbourhood analyses and cohort comparison

  Can be downloaded from https://github.com/biovault/ImaCytE

- Cytosplore (optional)

  Used for clustering and phenotype identification of large datasets

  Can be downloaded from www.cytosplore.org

- R studio

Step by step IMC analysis

Step 1| Conversion of .mcd files to .ome-tiff files in MCD viewer

1. Open the .mcd file and load one of the ROIs
2. Go to ‘File’ --> ‘export’
3. Change export type to <u>OME-TIFF 32-bit</u> and page to <u>multi</u>
4. Choose an export location in the ‘file name’ box (Figure 1a)
5. Select the ROIs to export and click ‘export’

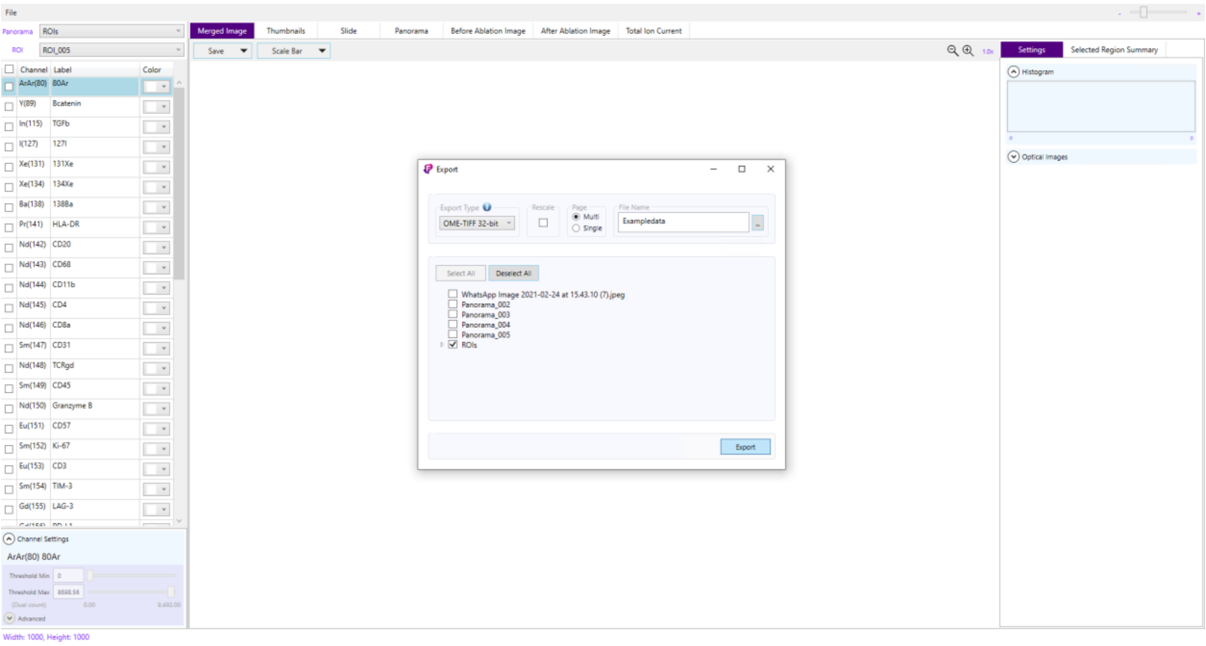


Figure 1a|Settings for export of .ome-tiff files

Step 2| Data normalisation with Penguin

1. Make sure all ROIs are located in one folder
2. Open anaconda prompt
3. Use the cd command to go to the IMC_preprocess_deploy folder on your PC. If this is on a different drive first go to the correct drive. For instance, if your data is on D:, type:

```
D:
```

4. Activate environment and launch jupyter the jupyter notebook

```
conda activate penguin
jupyter-notebook
```

5. A jupyter notebook opens and allows to choose between two notebooks:
   a. Check_th_all_ch_per_image.jpynb → use this notebook when using multi-TIFF files
6. Run the blocks of the jupyter notebook by clicking run for each block
7. After running the 'Parameter choice' block, change the path to the folder with your data and click 'Change path'
8. Now for each marker determine the optimal normalization threshold and percentile. In the channel dropdown select a channel and set the percentile and threshold. A good starting point is a threshold of 0.1 and percentile of 50.
9. In the 'compare images' tab the comparison between the raw image and the normalized image are shown.
10. When the optimal normalization values have been determined run the 'Saving options' block, add the same path as before and click 'Saving Options'
11. Fill in the table with the optimal normalization values.
12. In PathSave, add the path to save the images and click 'Save'.

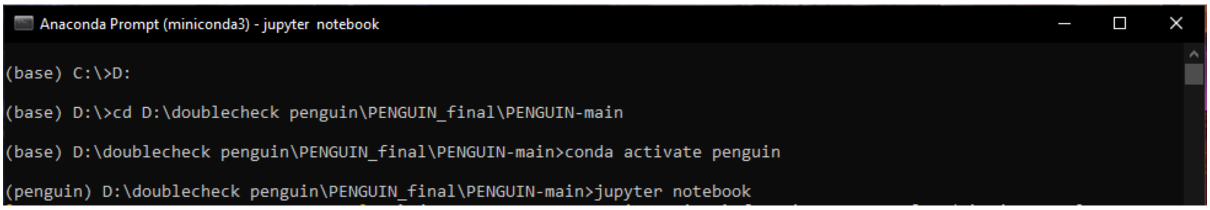


Figure 2a

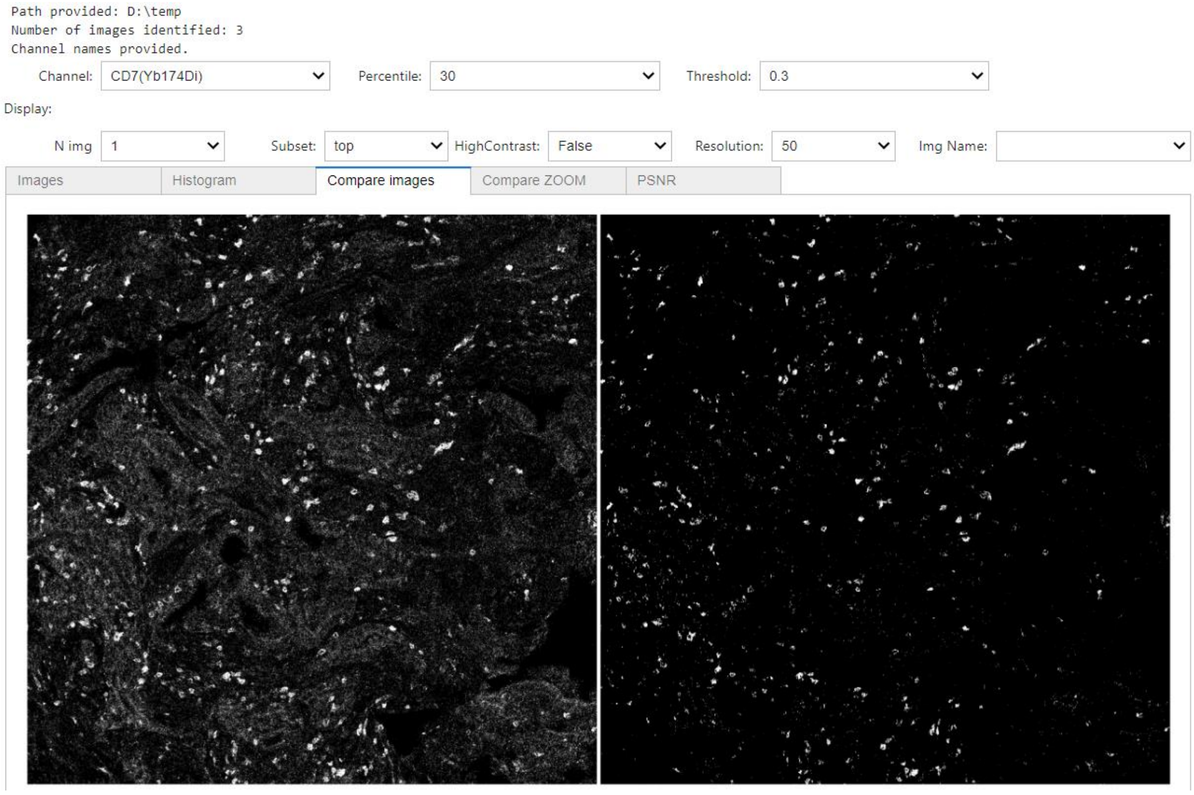


Figure 2b

index	threshold	percentile
80Ar(ArAr80Di)	0.1	50.0
Bcatenin(Y89Di)	0.1	50.0
TGFb(In115Di)	0.1	50.0
127I(I127Di)	0.1	50.0
131Xe(Xe131Di)	0.1	50.0
134Xe(Xe134Di)	0.1	50.0
138Ba(Ba138Di)	0.1	50.0
HLA-DR(Pr141Di)	0.1	50.0
CD20(Nd142Di)	0.1	50.0
CD68(Nd143Di)	0.1	50.0
CD11b(Nd144Di)	0.1	50.0
CD4(Nd145Di)	0.1	50.0
CD8a(Nd146Di)	0.1	50.0
CD31(Sm147Di)	0.1	50.0
TCRgd(Nd148Di)	0.1	50.0

Figure 2c

Step 3| Cell segmentation mask creation in Cellprofiler

To create cell segmentation masks, we perform sequential cell segmentation on lymphoid cells, myeloid cells, stroma cells, epithelial cells and remaining cells. These separate masks are combined again into one mask which is then exported. Therefore, before starting with the segmentation, determine what combination of markers cover the different cell types to include in a mask. Up to three markers can be selected per mask.

Typical marker combinations are:

Mask	Markers
Lymphoid	CD7, CD20, CD38
Myeloid	CD68, CD14, CD11c
Stromal	Vimentin
Epithelium	Keratin, B-catenin

1. Have the ‘cp_pipeline_2024’ pipeline ready
2. Start in the module ‘images’ in the bar on the left
3. Drag and drop all the folders generated in step 2 in the ‘ Drop files and folders here’ area
4. Go to the module ‘Metadata’ and click ‘Extract metadata’, followed by ‘Update
   Note: in the table, a frame should appear for each marker (in the example figure this is frame 1-10). If every image is only shown once, the metadata is not extracted properly. This can be solved by changing the ‘Metadata data type’ dropdown
5. Go to the module ‘NamesAndTypes. Here, the markers are assigned to use for cell segmentation. Make sure all markers that are needed for segmentation are annotated in this step. Click ‘Update’ when done.
6. While testing settings, run the pipeline in the ‘Test mode’, this allows checking the effect of settings and visual evaluation of the intermediate results. Clicking the eye next to a module determines if the results are shown or hidden.
7. If using different markers than standardly assigned, make sure to change this in the first steps of the script, in the corresponding ‘GrayToColor’ modules.
8. In the first ‘IdentifyPrimaryObjects’ module, the nuclei are identified using the DNA image. Changing the manual threshold or the threshold smoothing factor can help get the optimal segmentation. In the next steps, the nuclei are measured and very small cells are filtered out.
9. In the following blocks, the different membranes are added to the nuclei. First the lymphoid cells are segmented. If the membranes are not identified correctly, consider changing the manual threshold. Make sure that in both ‘IdentifyPrimaryObjects’ and

‘IdentifySecondaryObjects’ the same threshold is set. The lymphoid cells are automatically extracted and all remaining nuclei are taken along to the next steps.

10. Do the same for myeloid, stromal and epithelial cells. Changing the manual threshold to get the most optimal segmentation.
11. After determining the optimal settings, select the save folders for the cell_information files in the ‘ExportToSpreadsheet’ module
12. Create masks for all samples by clicking ‘Analyze images’ on the bottom of the program

    Note: by ticking the eyes next to the Analysis modules the software will show what each step does. However this is computational heavy and will crash slow computers.

13. Each masks is saved in the folder of each ROI, confirm that the masks are representative for the real data, otherwise some more tweaking of the CellProfiler project may be necessary.

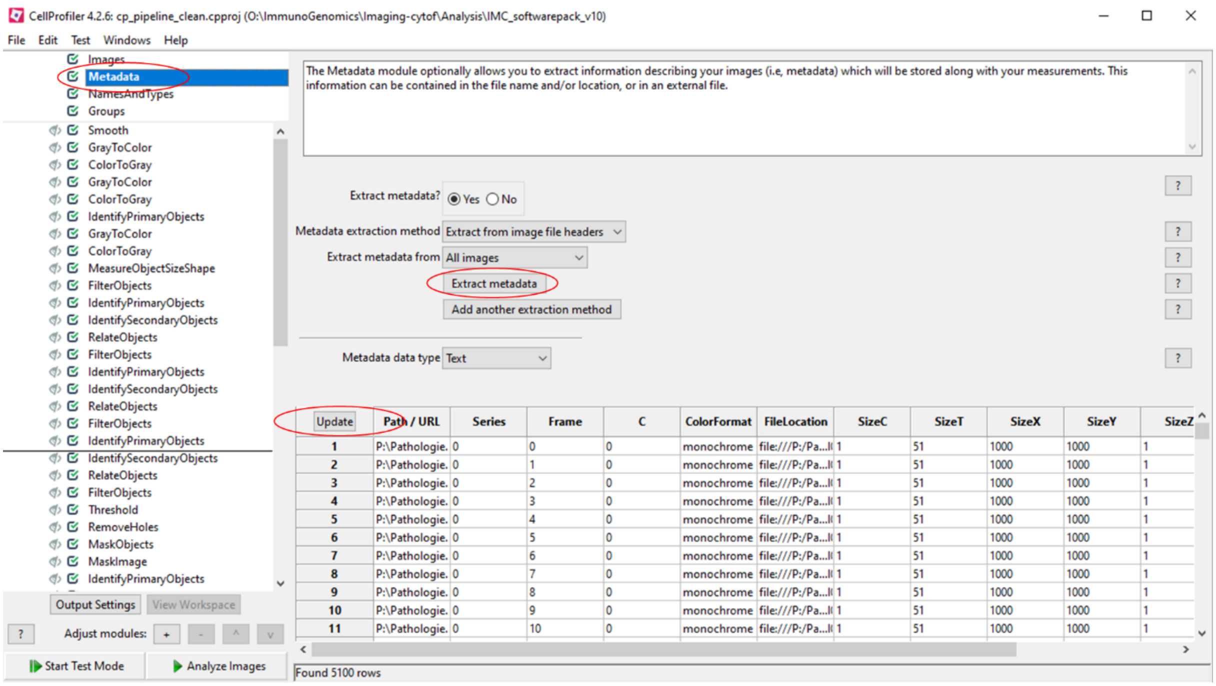


Figure 3a|

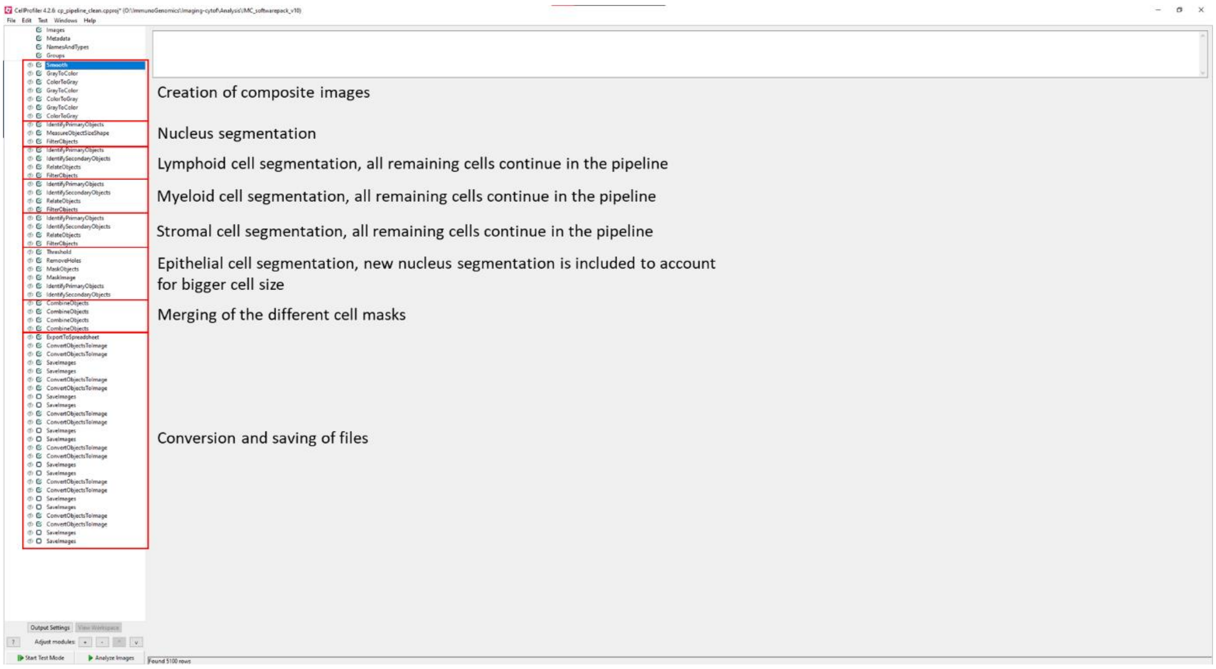


Figure 3b|

Step 4| Organisation of data to load into ImaCytE

4a| Using multi-TIFF files

1. For ImaCytE to load the data properly a folder is created for each ROI. If the manual is followed this is done automatically. Confirm in the next steps if all files are correct or create the folders manually.
2. Create a folder and name it "ROIs'
3. Place all folder containing the ROIs inside this folder (Figure 3)
4. Each ROI folder should contain (Figure 3):

   For unnormalized data analysis

   a. Ome.tiff file
   b. Mask file (uint16, name must end in _mask_1)

c. Tiles file

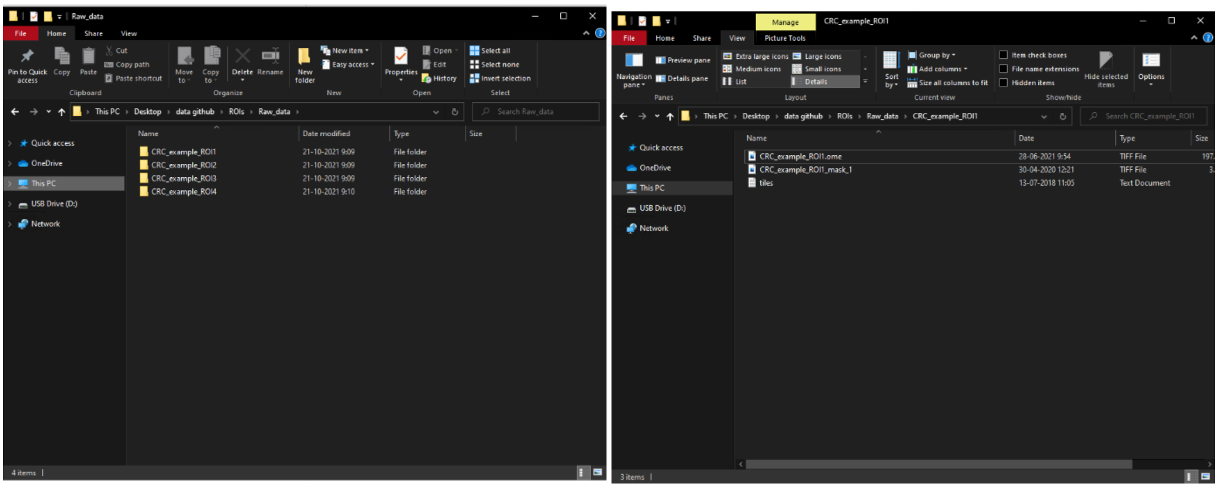


Figure 4a| Folder structure for ImaCyte

Step 5| Data visualization and inspection in ImaCytE

1. Run ImaCytE as administrator
2. Click ‘Options’ → ‘Load data’ → ‘Raw images’
3. Open the ‘ROIs’ folder and click ‘Select folder’
4. ImaCytE will show a loading bar and load the images
   Note: in rare cases the loading will crash and give an error. This is most likely due to the presence of the colour cell segmentation masks in the folders. Remove those and load again.
5. Visually inspect the images by selecting ‘samples’ and ‘markers’.
   Note: you can either choose several samples or markers, not both
6. You can now either run a TSNE analysis directly or export the single cell information via ‘Options’→ ‘export data per sample’ → ‘as fcs’ or ‘as .csv’ and perform phenotype identification with the method of choice

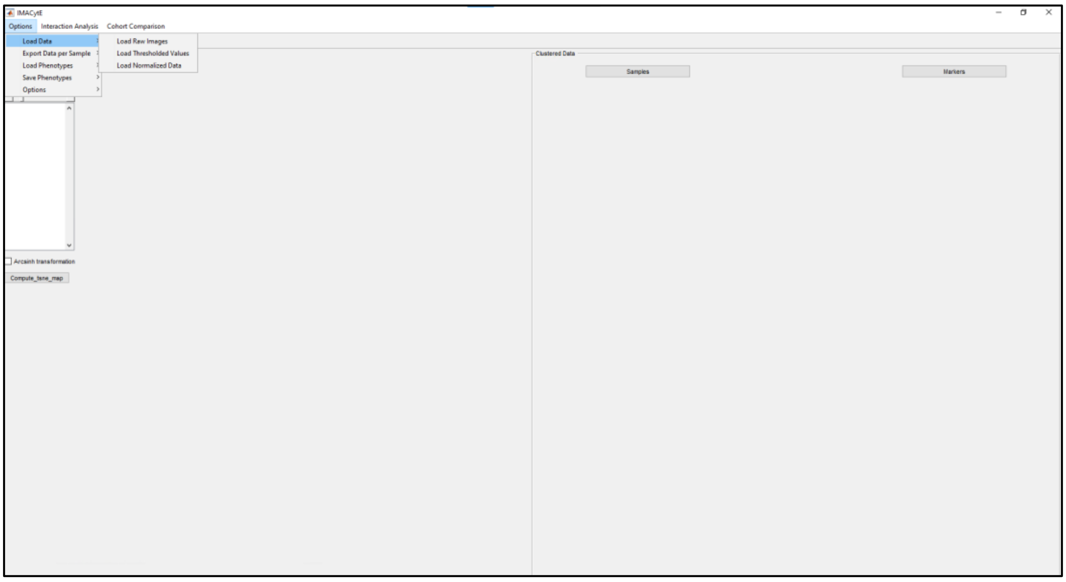


Figure 5a| ImaCytE: choose which data to load

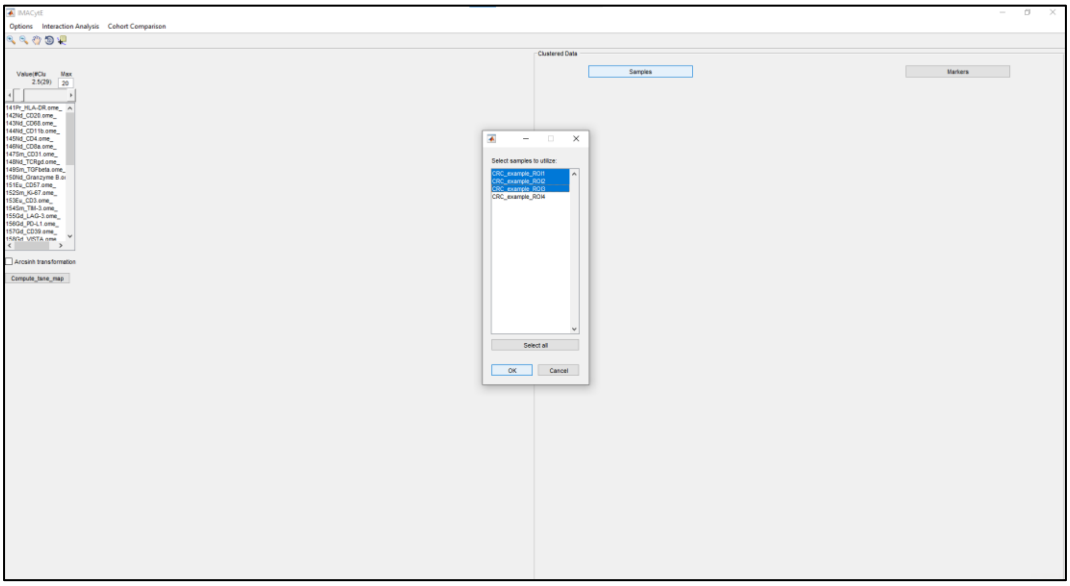


Figure 5b| ImaCytE: select images to visualize

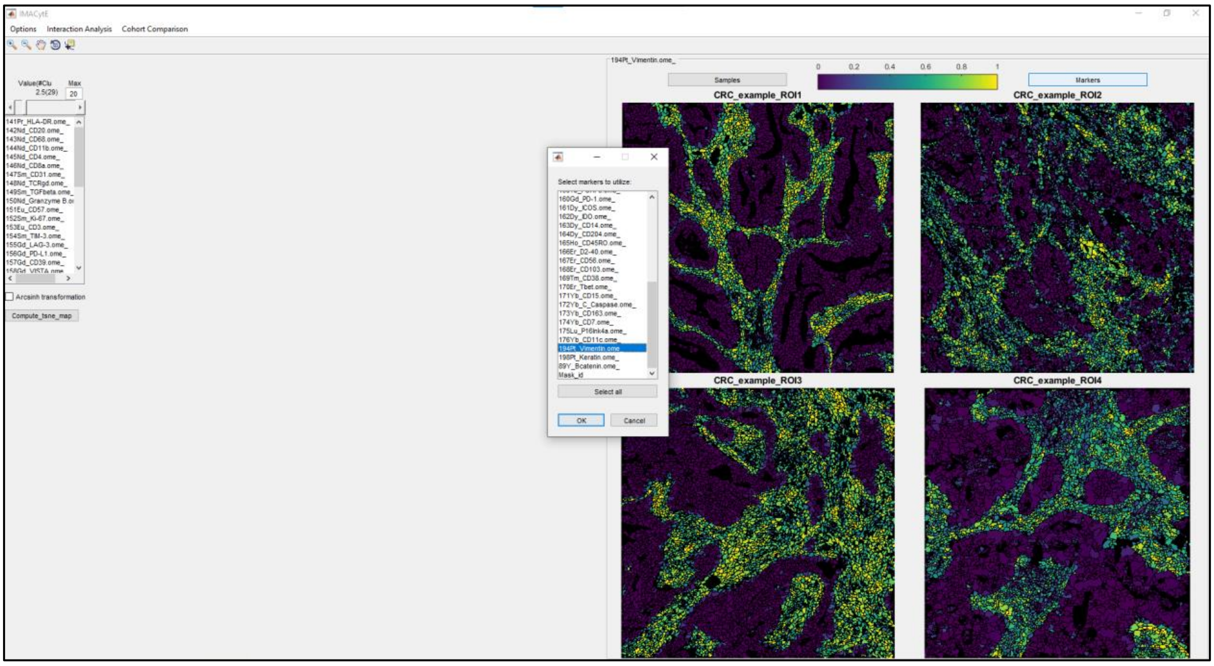


Figure 5c| ImaCytE: select markers to visualise

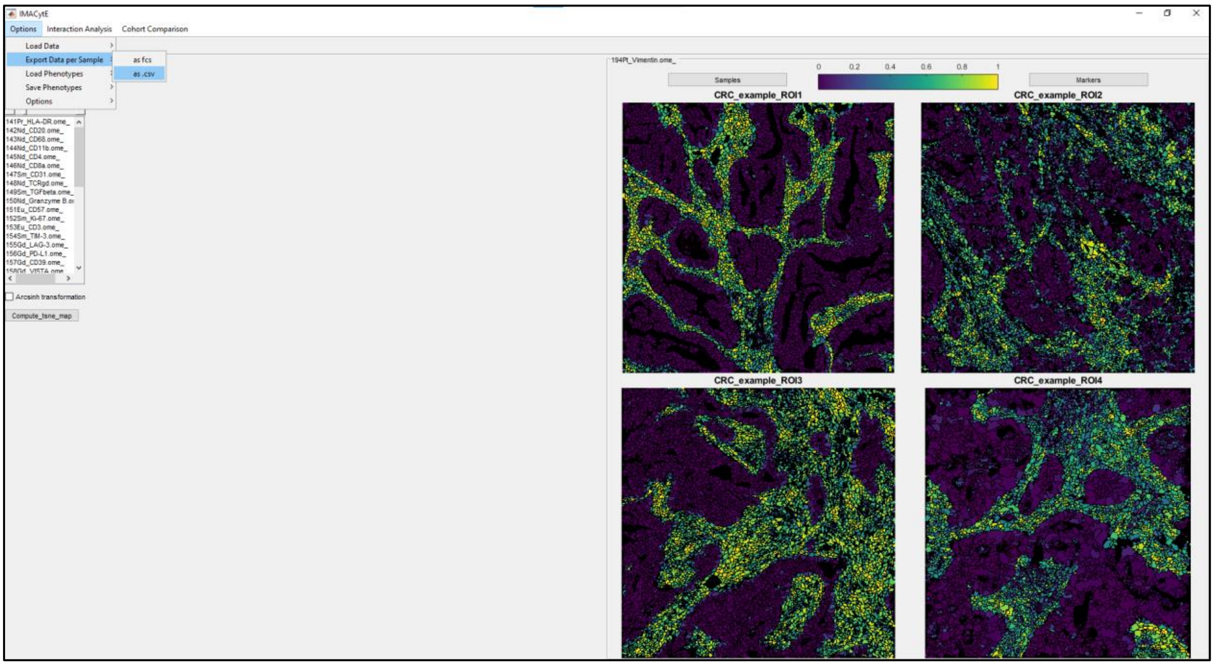


Figure 5d| Exporting single cell information as CSV or FCS

Step 6| Phenotyping through Cytosplore

1. In ImaCyte export the single cell data as .fcs files by clicking ‘Options’ and ‘Export data per sample’ and ‘as fcs’
2. Open Cytosplore, go to ‘file’ and ‘open… and select all single cell .fcs files you want to analyse.
3. A pop up screen will appear where you will select either TSNE (low complexity data) or HSNE (high complexity data)

   Note: Thresholded data is normalized and thus ArcSin Transform is redundant.
4. In the ‘Active marker’ panel, select all markers to be used for clustering (figure 18)

   Note: excluded markers can still be visualized, but are not clusterparameters
5. Select the number of scales to use for analysis. 2 scales works well for 25-100 images
6. Go to ‘Advanced’ in the drop down menu and choose the number of iterations to run. 5000 works well for most datasets
7. ‘Compute HSNE’, after computing the first scale there are a number of options (figure 19):
   a. Data is very scattered → select all datapoints, rightclick ‘zoom into selection’ and continue with the next step
   b. Data is very dense → splitting data in tumourcells and stromal cells will improve phenotyping
   c. Data is the right density → continue with the next step
8. In the drop down ‘sample’ you can go through the markers and see if the markers are clustered together
9. To cluster go to the drop down ‘Single-cell’ and select ‘Clustering’ (figure 20)
10. Change the sigma and press ‘cluster’ so that all islands are separated

    Note: you can later merge clusters, so it is useful to oversegment clusters slightly
11. Go to ‘View’ and ‘Heatmap View’ this will pop up a second window where under ‘clusters’ you can select the clusters of this analysis (figure 21)
12. By rightclicking the markers you can ‘toggle marker selection’ to remove any channels that are not informative. Click ‘toggle marker selection’ again to save (figure 22)
13. Toggling the dendogram orders all clusters by similarity. Together with the HSNE/TSNE map you can now merge clusters that are similar (figure 23)
14. Shift-click clusters to be merged and right click, then click ‘merge selected clusters’

    Note: merging of clusters cannot be undone
15. After merging similar clusters you can save the phenotypes by rightclicking the heatmap. Make sure to save everything: ‘Save clusters as FCS’, ‘Save clusters (per sample) as FCS’, ‘Save Heatmap as image’, ‘Save Heatmap as CSV’.

16. Export the TSNE plots by clicking the camera on the right bottom of Cytosplore, select all markers to be exported, select a folder to export to and click ‘capture’ (figure 24)

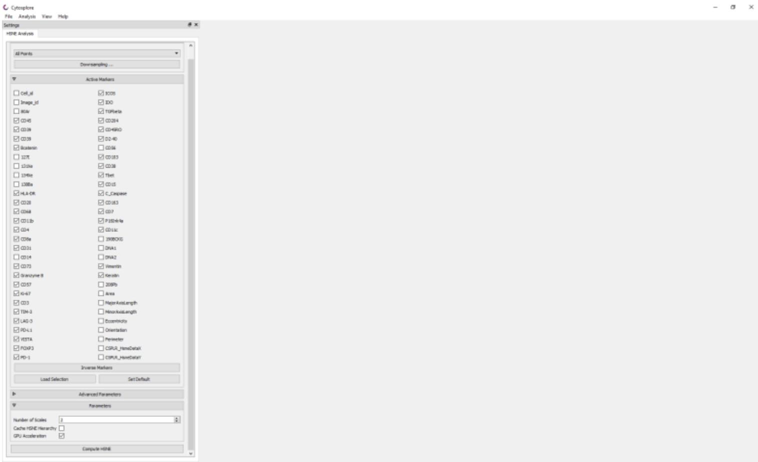


Figure 6a| CytoSplore: marker selection for TSNE/HSNE

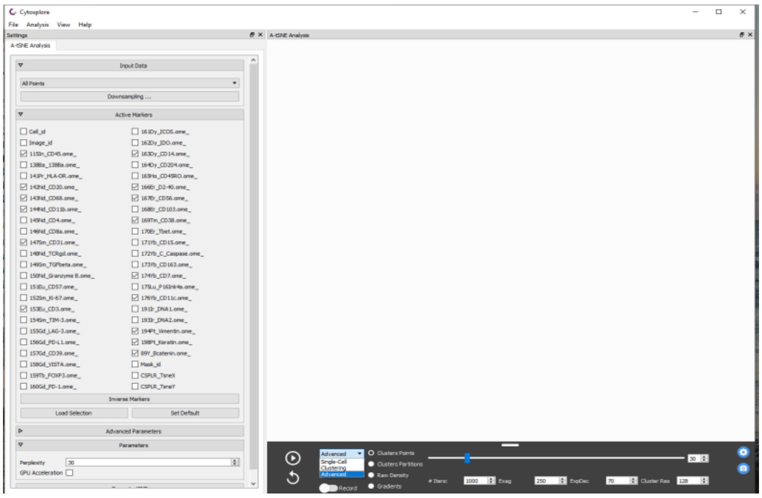


Figure 6b|Cytosplore: change iterations to run

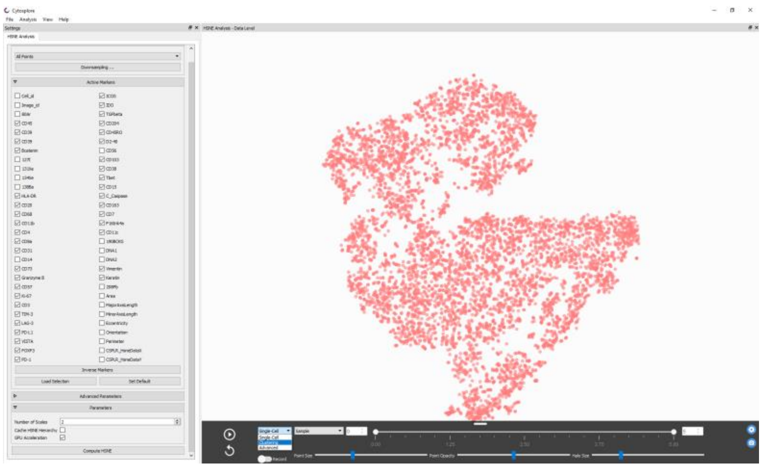


Figure 6c| CytoSplore: TSNE computation and selection of markers to visualise

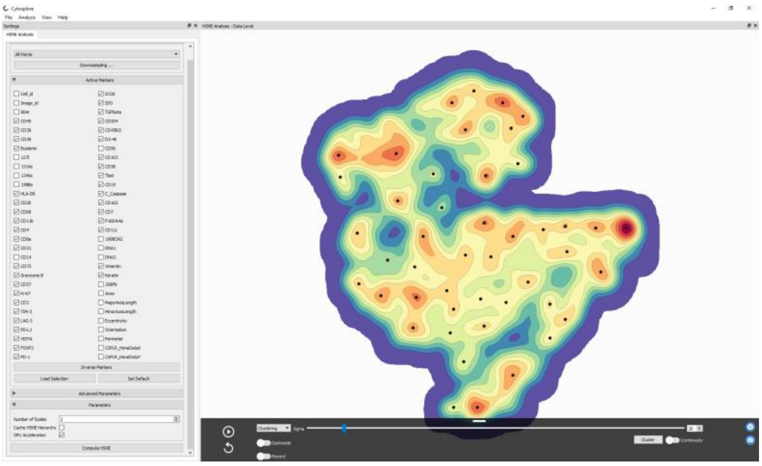


Figure 6d| CytoSplore: clustering depth

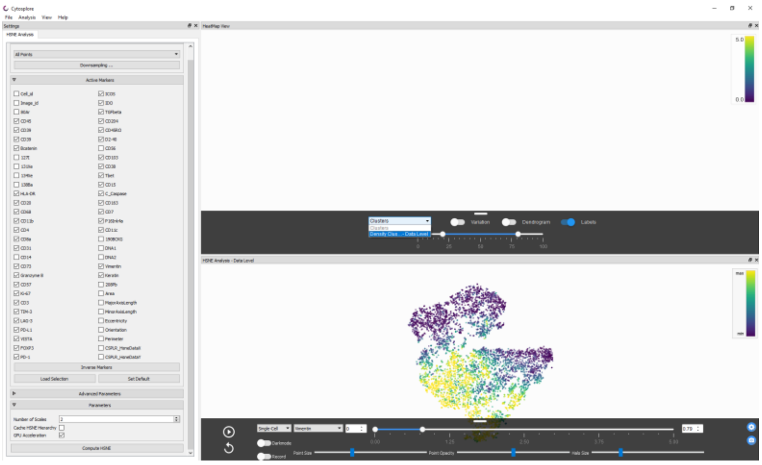


Figure 6e| CytoSplore: Heatmap and TSNE view

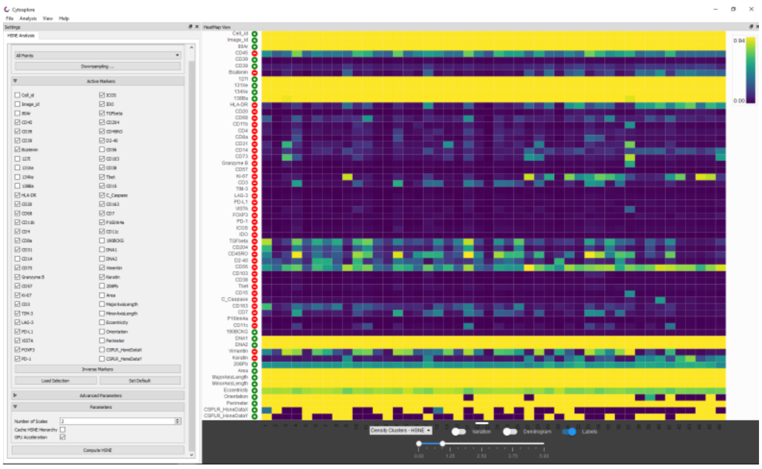


Figure 6f| CytoSplore: Heatmap with clusters

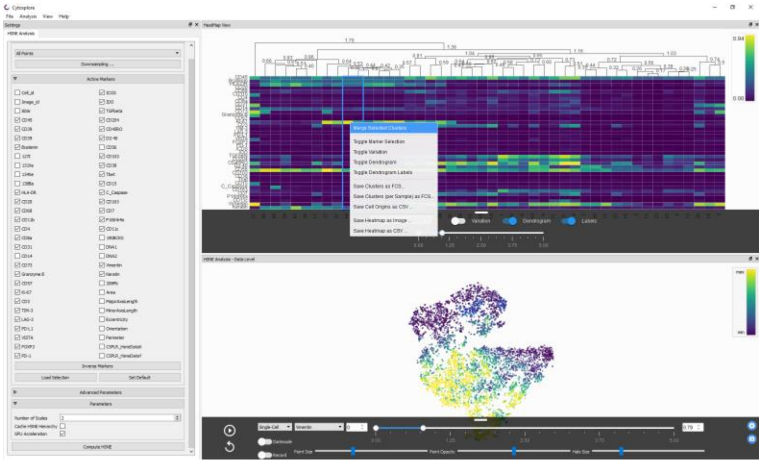


Figure 6g| CytoSplore: merging of clusters with similar marker profiles

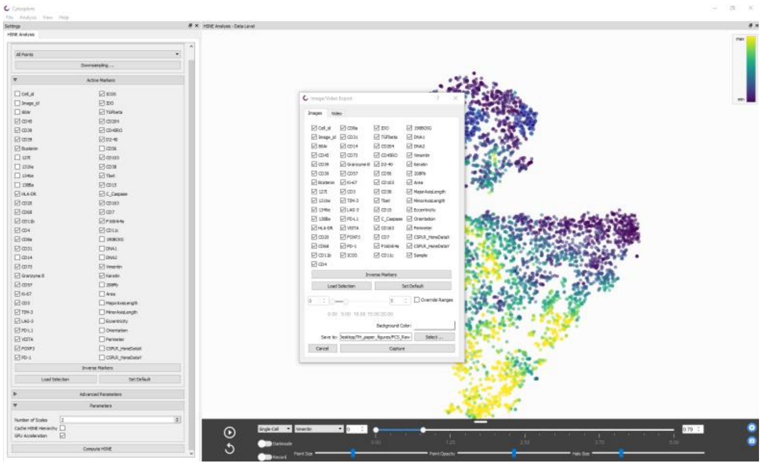


Figure 6h| CytoSplore: Exporting of TSNE maps

Step 7| Loading data back into ImaCytE

1. Load the images as explained in step 5
2. Phenotype FCS files can be reloaded into ImaCytE via 'Options' → 'Load phenotypes' → 'per phenotype (.FCS)'
3. After reloading, FCS files can be merged further if needed and the session containing these can be saved, which can be reloaded.
4. Visual inspection of the clusters can be done by clicking each cluster to determine where in the images these occur. Make sure to check those back on the original images in MCD viewer.
5. Further merging of clusters can be done by selecting both and right-clicking → merge clusters
6. Colours of clusters can be changed by double clicking the colour coding below the cluster
7. Final clusters can be exported for subsequent analysis and reporting by via 'Options' → 'Save phenotypes' → 'Per sample'

8. To save the session with merged phenotypes for later use, click 'Options' → 'Save phenotypes' → 'Per sample'

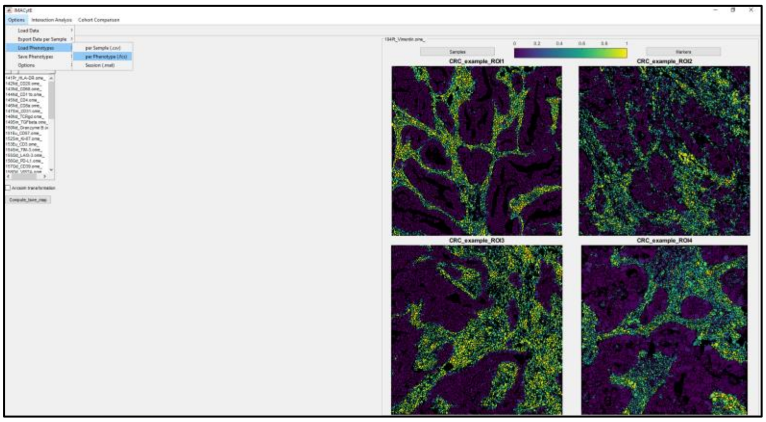


Figure 7a| importing phenotype files (.fcs)

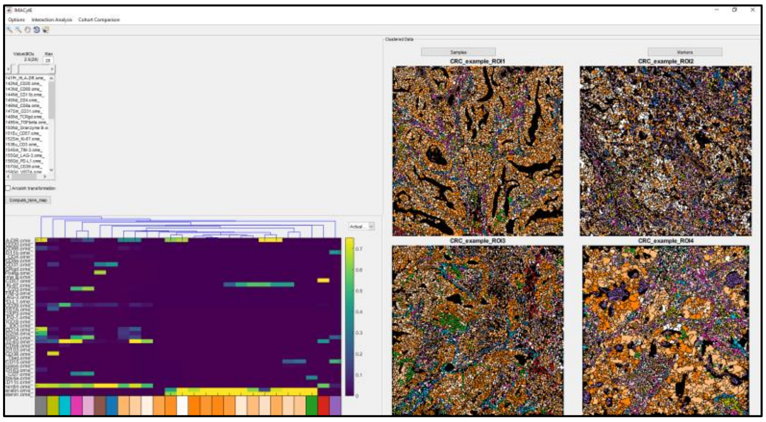


Figure 7b| the phenotypes will be put into a heatmap and mapped onto the images

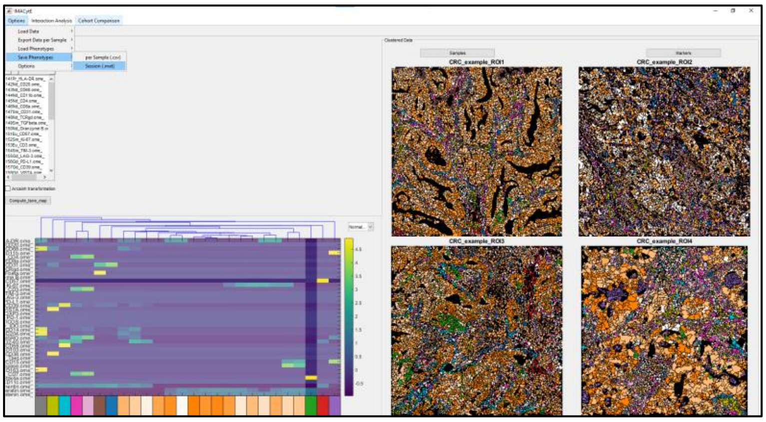


Figure 7c| Saving the heatmap/phenotype session

Step 8| Reporting of results

For the downstream visualisation and analysis of results, we created a number of R-scripts to utilise:

Script	Used for
1_File_merging_namematch	To combine phenotypes, intensities, cell coordinates and metadata in a large data frame for subsequent analysis
2_Phenotype_Counting	To count the abundance of each phenotype per image and sample
3_Marker_expression_heatmap	To create a heatmap of marker expression per phenotype
4_Phenotype_abundance_heatmap	To create a heatmap with the abundance of each phenotype per sample
5_Phenotype_abundance_graphs	To create graphs for each phenotype and their abundance between groups of samples based on provided metadata
6_Intensity_counting	To count the abundance of each phenotype positive for markers of interest
7_Markerpositivity_graphs	To create graphs of the counts generated in script 6
8_Phenotype_image_visualisation	To visualise the phenotypes onto a selected image
8a_Phenotype_image_visualisation_all_ROIs	To visualise the phenotypes on all images
9_Spatial_neighbour_detection	To identify cell-cell neighbourhoods
9a_Spatial_neighbourhood_plots	To plot cell-cell neighbourhood abundances
10_Spatial_neighbourhood_hubs	To identify multicellular neighbourhoods and plot those

<u>1_File_merging_namematch</u>

Input files:

- phenotype files per sample (from step 7.7)
- intensity files per sample (from step 5.6)
- cell metrics csv: (from step 3.11)
- metadata csv: containing an id column with the ROI name, a sample column which contains the patient name for each ROI and one of more group/variable columns

Usage: run the script

Output: A dataframe containing all cells and matched coordinates, phenotypes, intensities and metadata. As well as a metadata data frame

	A	B	C	D
1	id	sample	group	
2	ESD1_01	ESD1	Carcinoma	
3	ESD1_02	ESD1	High grade dysplasia	
4	ESD1_03	ESD1	Carcinoma	
5	ESD1_04	ESD1	High grade dysplasia	
6	ESD1_05	ESD1	High grade dysplasia	
7	ESD1_06	ESD1	Carcinoma	
8	ESD1_07	ESD1	Normal	
9	ESD1_08	ESD1	Transition	
10	ESD1_09	ESD1	Normal	
11	ESD2_01	ESD2	Carcinoma	
12	ESD2_02	ESD2	Low grade dysplasia	
13	ESD2_03	ESD2	Low grade dysplasia	
14	ESD2_04	ESD2	Low grade dysplasia	
15	ESD2_05	ESD2	Carcinoma	
16	ESD3_01	ESD3	Normal	
17	ESD3_02	ESD3	Low grade dysplasia	
18	ESD3_03	ESD3	Normal	
19	ESD3_04	ESD3	Low grade dysplasia	

Figure 8a| metadata example

2_Variables

Input files:

- Cell_matrix.Rdata
- Metadata_sample.Rdata

Usage: change all variables marked with comments and run

Output: Variables that are required for downstream plotting and analyses

3_Phenotype_Counting

Input files:

- Cell_matrix.Rdata
- Metadata.Rdata

Usage: run full script, ensure variables from 2_Variables.R are loaded

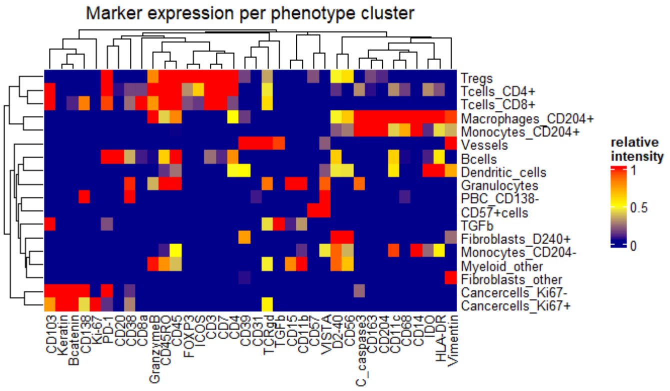
Output: Dataframes and backup CSV files containing phenotype and category counts for all images and samples

4_Marker_expression_heatmap

Input files:

- Cell_matrix.Rdata
- Metadata_sample.Rdata

Usage: run full script, ensure variables from 2_Variables.R are loaded. What markers are shown can be changed in the 2_Variables.R script



Output: heatmap with marker expression per phenotype cluster

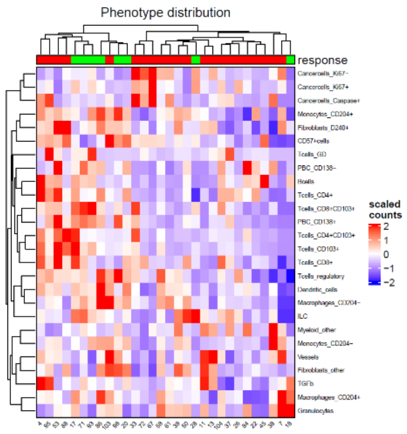
5_Phenotype_abundance_heatmap

Input files:

- catcounts_sample.Rdata (from script 3)

Usage: run full script, ensure variables from 2_Variables.R are loaded. What metadata is used for annotation and its colours can be changed in the 2_Variables.R script.

Output: heatmap with phenotype abundances per sample



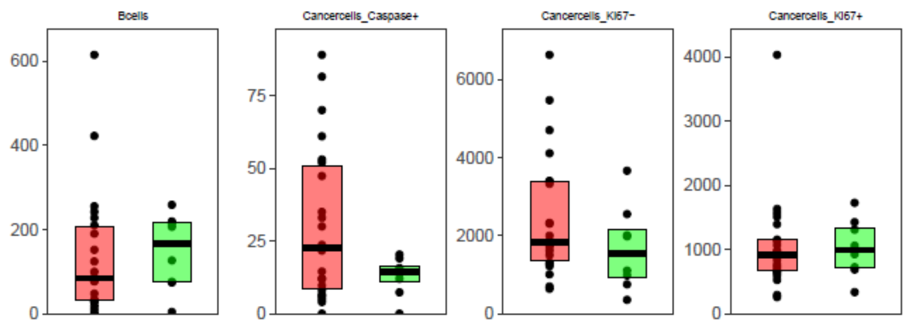
6_Phenotype_abundance_graphs

Input files:

- catcounts_sample.Rdata (from script 3)

Usage: run full script, ensure variables from 2_Variables.R are loaded. A pheno_order or different colours can be assigned in the 2_Variables.R script.

Output: graphs with phenotype abundances per sample



7_Intensity_counting

Input files:

- Cell_matrix.Rdata
- catcounts_sample.Rdata (from script 3)
- Metadata_sample.Rdata
- Metadata.Rdata

Usage: run full script, ensure variables from 2_Variables.R are loaded. What markers are analysed can be changed in the 2_Variables.R script.

Output: Dataframes and backup CSV files containing number of cells of each phenotype above 0.2 intensity threshold for all markers

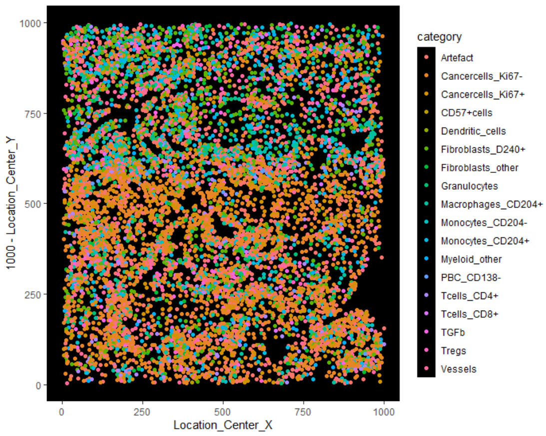
9_Phenotype_image_visualisation

Input files:

- Cell_matrix.Rdata

Usage: change the ‘image_to_plot’ variable to the image of interest and run full script

Output: Image with xy coordinates of all cells and their phenotypes



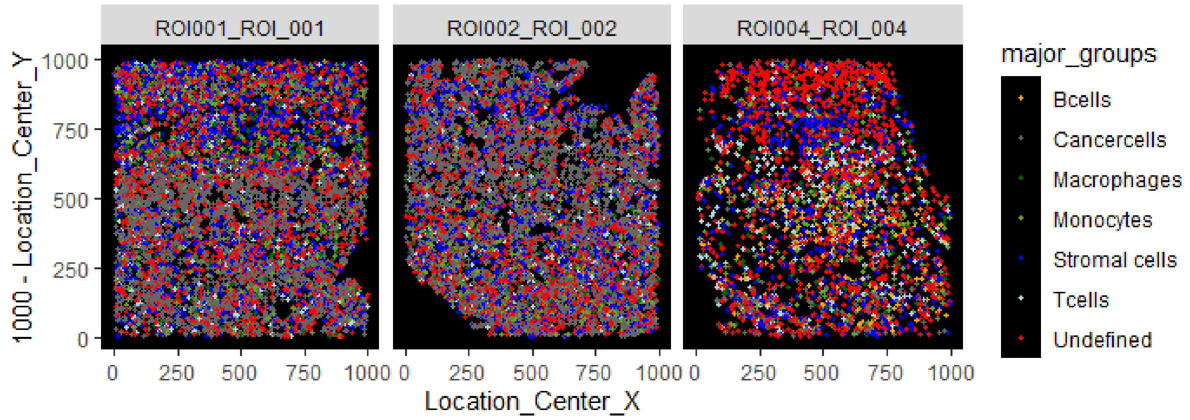
9a_Phenotype_image_visualisation_all_ROIs

Input files:

- Cell_matrix.Rdata

Usage: run full script, ensure variables from 2_Variables.R are loaded. Major lineages can be assigned/adapted in the 2_Variables.R script.

Output: pdf with images containing the xy coordinates and their major lineages/phenotypes of all ROIs



10_Spatial_neighbour_detection

Input files:

- Cell_matrix.Rdata
- Metadata.Rdata

Usage: run full script, ensure variables from 2_Variables.R are loaded. In the 2_Variables.R script, you can choose for knn or expansion and what distances are assessed.

Output: Dataframe with either expansion or knn neighbourhood counts and significance testing

11_Spatial_neighbourhood_plots

Input files:

- Cell_matrix.Rdata
- Metadata.Rdata

Usage: run full script, ensure variables from 2_Variables.R are loaded

Output: Heatmap with relative number of interactions per cell and significance